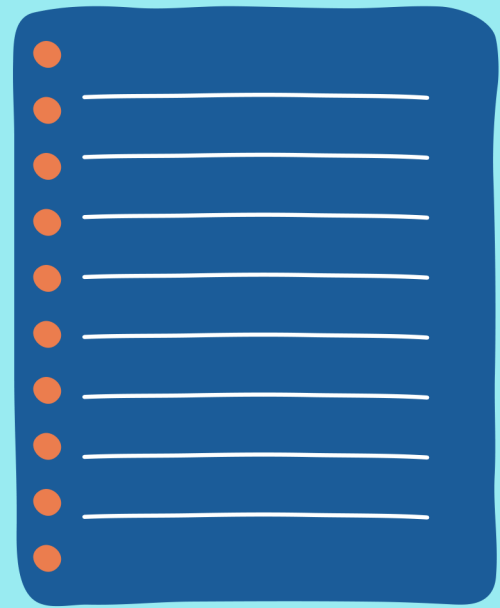
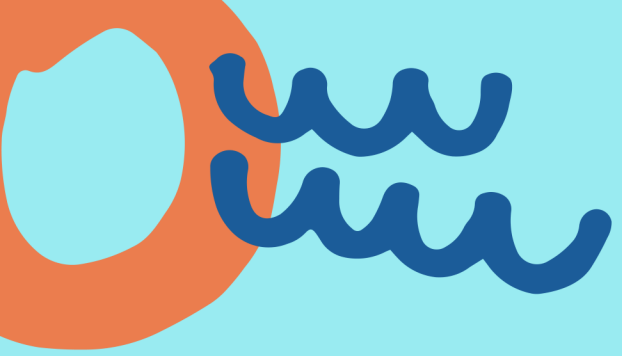


GOLANG DB TRANSACTION

Backend Master Class

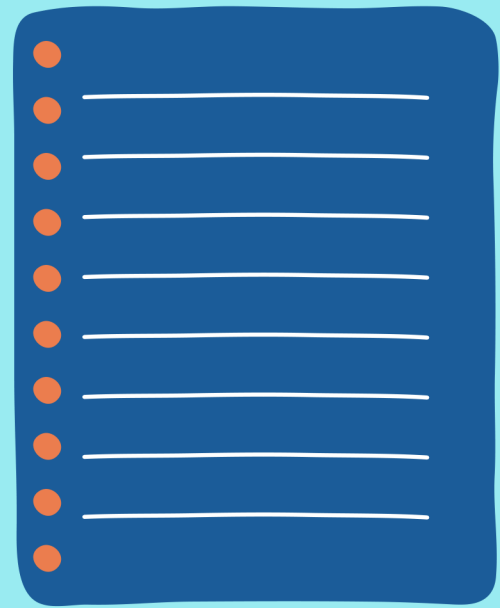
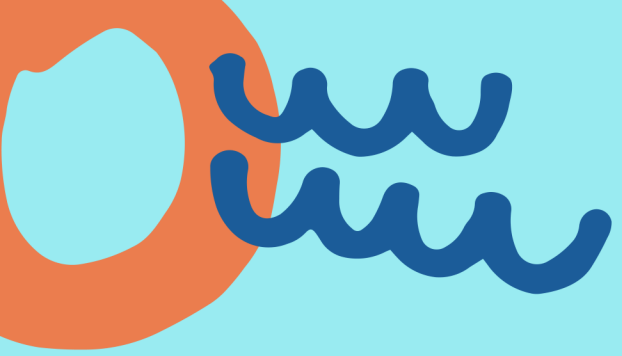
LECTURE 6





What is a db
transaction?

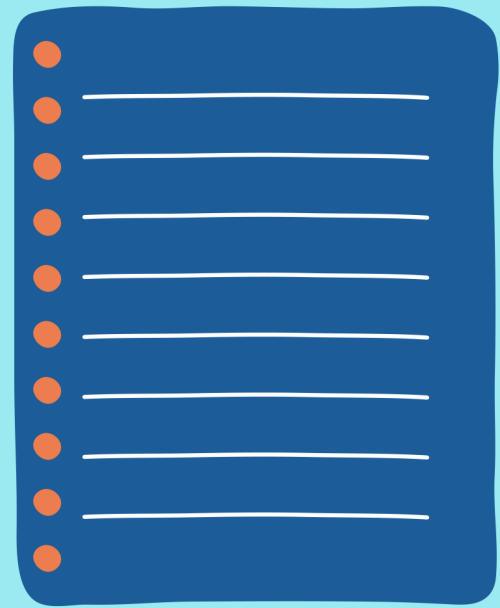
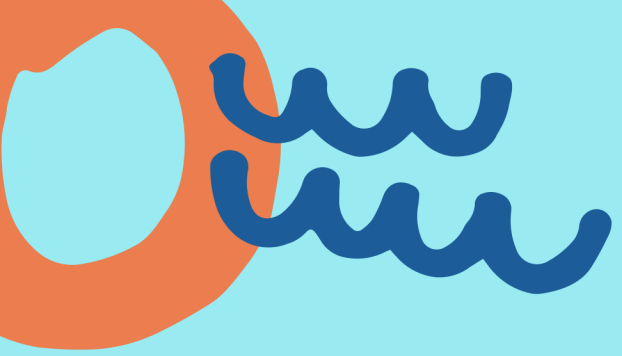




What is a db transaction?

- A single unit of work





What is a db transaction?

- A single unit of work
- Often made up of multiple db operations



EXAMPLE

xx

EXAMPLE

Transfer 10 USD
from bank account 1
to bank account 2

XX

EXAMPLE

Transfer 10 USD
from bank account 1
to bank account 2

XX

1

Create a transfer record with
amount = 10

EXAMPLE

Transfer 10 USD
from bank account 1
to bank account 2

XX

1

Create a transfer record with
amount = 10

2

Create an account entry for
account 1 with amount = -10

EXAMPLE

Transfer 10 USD
from bank account 1
to bank account 2



- 1 Create a transfer record with amount = 10
- 2 Create an account entry for account 1 with amount = -10
- 3 Create an account entry for account 2 with amount = +10

EXAMPLE

Transfer 10 USD
from bank account 1
to bank account 2



- 1 Create a transfer record with amount = 10
- 2 Create an account entry for account 1 with amount = -10
- 3 Create an account entry for account 2 with amount = +10
- 4 Subtract 10 from the balance of account 1

EXAMPLE

Transfer 10 USD
from bank account 1
to bank account 2



- 1 Create a transfer record with amount = 10
- 2 Create an account entry for account 1 with amount = -10
- 3 Create an account entry for account 2 with amount = +10
- 4 Subtract 10 from the balance of account 1
- 5 Add 10 to the balance of account 2

Why do we need db
transaction?



Why do we need db transaction?

1

To provide a reliable and consistent unit of work, even in case of system failure



Why do we need db transaction?

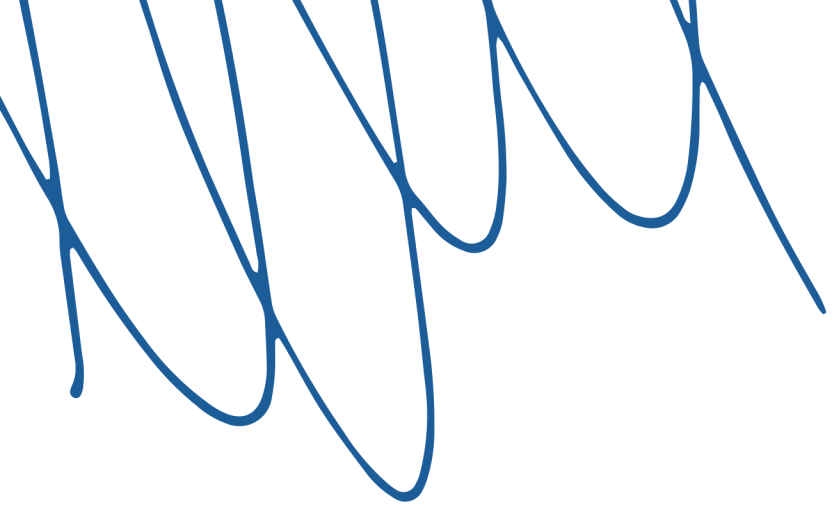
1

To provide a reliable and consistent unit of work, even in case of system failure

2

To provide isolation between programs that access the database concurrently





ACID PROPERTY





1

Atomicity (A)

Either all operations complete successfully or the transaction fails and the db is unchanged.

ACID PROPERTY





ACID PROPERTY

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Atomicity (A)

Either all operations complete successfully or the transaction fails and the db is unchanged.

2

Consistency (C)

The db state must be valid after the transaction. All constraints must be satisfied.





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Concurrent transactions must not affect each other.





ACID PROPERTY

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Either all operations complete successfully or the transaction fails and the db is unchanged.

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Consistency (C)

The db state must be valid after the transaction. All constraints must be satisfied.

3

Isolation (I)

Concurrent transactions must not affect each other.

4

Durability (D)

Data written by a successful transaction must be recorded in persistent storage



HOW TO RUN SQL TX?



HOW TO RUN SQL TX?

```
BEGIN;
```

```
...
```

```
COMMIT;
```



HOW TO RUN SQL TX?

```
BEGIN;
```

```
...
```

```
COMMIT;
```

```
BEGIN;
```

```
...
```

```
ROLLBACK;
```

